CLINICAL · NOSE & SINUSES

# Fungal sinusitis

Fungi in the sinuses cause two utterly different diseases. In a healthy host they sit in a sinus as a ball or provoke an allergic storm of polyps and thick mucin, and surgery cures or controls them. In a diabetic in ketoacidosis or a neutropenic patient they invade blood vessels, turn the turbinate black and can kill within days. The first question is always whether the fungus is invading tissue.

**AT A GLANCE**

- Invasive disease is the emergency.
- Biopsy a pale turbinate in the immunocompromised.
- Antifungals, debridement, reverse the cause.
- Non-invasive disease needs clearance, not antifungals.

## 1 CHAPTER ONE Overview

Invasive or not: everything follows.

### 1.1 Classification

**FUNGAL RHINOSINUSITIS**

- NON-INVASIVE **Fungal ball (mycetoma)** a mass in one sinus, healthy host
- NON-INVASIVE **Allergic fungal rhinosinusitis** hypersensitivity, polyps, eosinophilic mucin
- NON-INVASIVE **Saprophytic infestation** fungal growth on crusts after surgery
- INVASIVE **Acute fulminant** immunocompromised; Mucor, Aspergillus; days
- INVASIVE **Chronic invasive** slow, months; often diabetic
- INVASIVE **Granulomatous invasive** immunocompetent; Aspergillus flavus; Sudan, India, Pakistan

## 1.2 Organisms

*Aspergillus* (fumigatus, flavus, niger), the Mucorales (*Mucor*, *Rhizopus*), dematiaceous fungi (*Alternaria*, *Bipolaris*, *Curvularia*) and, rarely, *Candida*, *Histoplasma* and *Blastomyces*.

## 1.3 Mucor or Aspergillus?

MUCOR OR ASPERGILLUS?

| | MUCOR | ASPERGILLUS |
|---|---|---|
| **Hyphae** | Broad | Narrow |
| **Septa** | Rarely septate | Always septate |
| **Branching** | At right angles | At acute angles, about 45 degrees |
| **Typical disease** | Rhino-orbito-cerebral invasion, black necrosis | Fungal ball, allergic disease, chronic invasive disease |

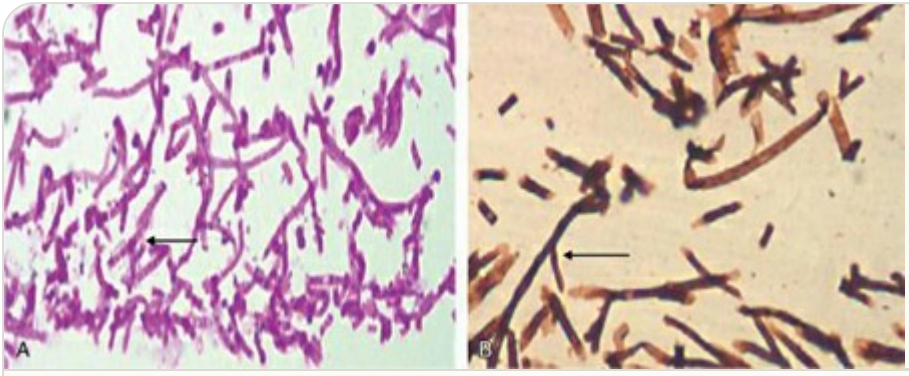



**FIGURE 1** Aspergillus: septate hyphae branching at acute angles (arrow), on H&E and silver stain. Dhingra, fig. 37.3

# 2 CHAPTER TWO Fungal ball

A mass in a healthy sinus.

## 2.1 Picture

A dense, cheesy or greyish-black mass of fungal hyphae, usually in one maxillary sinus (then sphenoid, ethmoid, frontal), in an otherwise healthy, often older patient. Symptoms are those of one-sided chronic sinusitis, or none.

## 2.2 CT

A single opacified sinus with a hyperdense centre or calcific flecks, and no bone erosion; the wall may be thickened.

## 2.3 Treatment

Endoscopic removal of the ball and a wide opening of the sinus. No antifungal drugs are needed.

# 3 CHAPTER THREE Allergic fungal rhinosinusitis

An allergic storm.

## 3.1 What it is

A hypersensitivity (type I and III) reaction to fungi growing in the sinus mucin, without tissue invasion, in immunocompetent, often young, atopic patients; commoner in warm, humid climates.

## 3.2 Bent and Kuhn criteria

- Type I hypersensitivity to fungi.

- Nasal polyposis.
- Characteristic CT findings.
- Eosinophilic mucin without fungal invasion of tissue.
- Positive fungal stain of the mucin.

## 3.3 Clinical picture

Gradual, often one-sided polyposis, thick, dark, peanut-butter mucin, and slow expansion of the sinuses that can displace the eye (proptosis, telecanthus) or erode the skull base. The mucin contains eosinophils, *Charcot-Leyden crystals* and scattered hyphae.



**FIGURE 2** Proptosis and lateral displacement of the eye from allergic fungal polyposis.
Dhingra, fig. 41.2

## 3.4 Imaging

CT shows expanded, opacified sinuses with heterogeneous high-density material (the *double density* sign), thinning and remodelling of bone. On MRI the mucin is dark on T2.

### 3.5 Treatment

Complete endoscopic clearance of mucin and polyps with wide drainage, combined with oral steroids before and after surgery, long-term topical steroids and saline irrigation, and close follow-up because recurrence is common. Immunotherapy and antifungals have a limited role.

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CHAPTER FOUR

# 4 Acute invasive fungal sinusitis

A surgical and medical emergency.

### 4.1 Who gets it

Patients with uncontrolled diabetes, particularly with ketoacidosis (the acid environment favours Mucor), neutropenia, leukaemia, chemotherapy, organ transplants, HIV, and high-dose steroids. After the second wave of COVID-19, thousands of cases of rhino-orbito-cerebral mucormycosis, popularly called *black fungus*, followed steroid treatment in diabetics.

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### 4.2 Mechanism

The hyphae invade arteries, causing thrombosis and ischaemic necrosis, and spread rapidly along vessels and nerves to the orbit, cribriform plate, cavernous sinus and brain.

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### 4.3 Clinical picture

Fever, facial pain or numbness, nasal blockage and bloodstained discharge; then lid swelling, proptosis, ophthalmoplegia and loss of vision; palatal or facial skin discolouration; confusion. Endoscopy shows pale, insensate mucosa early, then *black necrotic eschar* on the turbinate, septum or palate.




**FIGURE 3** Mucormycosis: (A) slight swelling of the right lower lid, (B) dark discolouration of the right palate and alveolus, (C) black necrotic tissue in the right nasal cavity.

Dhingra, fig. 28.4




**FIGURE 4** Mucormycosis with cheek and orbital involvement, and destruction of the palate.

Dhingra, fig. 28.5

## 4.4 Diagnosis

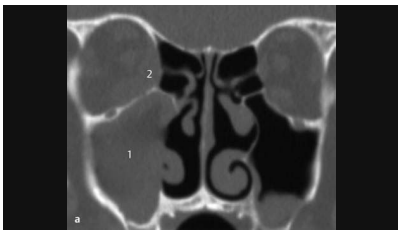
Urgent endoscopic biopsy of the middle turbinate or suspicious mucosa, with frozen section and fungal stains (silver, PAS), and culture. Contrast CT and MRI show the extent of soft tissue, orbital and intracranial spread; MRI may show non-enhancing, devitalised mucosa.

**IMAGING IN THE APP** 2 cases

Non-invasive and invasive fungal sinusitis.



Noninvasive Fungal Sinusitis
Infection & inflammation · 9 images



Invasive Fungal Sinusitis
Infection & inflammation · 10 images

## 4.5 Treatment

**SUSPECTED INVASIVE FUNGAL SINUSITIS**

1. Urgent endoscopy and biopsy with frozen section.
2. Start liposomal amphotericin B (about 5 mg/kg/day) immediately; monitor renal function.
3. Correct ketoacidosis and glucose; stop or reduce immunosuppression; support neutropenia.
4. Urgent surgical debridement back to bleeding tissue, endoscopic or open, repeated as needed.
5. Involve ophthalmology, infectious diseases and the medical team.

- **Antifungals:** liposomal amphotericin B for Mucor; posaconazole or isavuconazole for step-down or intolerance; voriconazole for invasive *Aspergillus* (ineffective against Mucor).
- **Surgery:** debridement may extend to medial or total maxillectomy and orbital exenteration in extensive disease, with later obturator rehabilitation of the palate.

## 4.6 Outcome

Mortality is high, especially with intracranial spread; survival depends on reversing the immune defect and early, complete debridement.

**OPERATIVE GUIDES IN THEATRE** 3 operations

**Endoscopic Ethmoidectomy**
Endoscopic debridement
**Medial Maxillectomy**
Medial maxillectomy
**Complete Maxillectomy**
Maxillectomy in extensive disease

# 5 CHAPTER FIVE Chronic invasive fungal sinusitis

Slow invasion.

## 5.1 Chronic invasive

Slowly progressive invasion over months, often in diabetics or patients on long-term steroids, presenting as chronic sinusitis with bone erosion and orbital apex syndrome or skull base spread. Biopsy shows fungal invasion of the submucosa. Surgical debridement and long-term antifungals.

## 5.2 Granulomatous invasive

In immunocompetent patients in Sudan, India and Pakistan, usually *Aspergillus flavus*: a slowly enlarging mass of the cheek, orbit or nose with non-caseating granulomas and giant cells. Surgery and long-term azoles.

# 6 CHAPTER SIX Differential diagnosis

Mimics.

- Bacterial chronic rhinosinusitis with polyps.
- Inverted papilloma and sinonasal malignancy (one-sided mass with bone erosion).
- Granulomatosis with polyangiitis and NK/T-cell lymphoma (destructive, crusted nose).
- Mucocele (expansion without the hyperdense mucin).

# 7 CHAPTER SEVEN For the examination

What the examiner wants to hear.

## 7.1 Viva essentials

**WHAT THE EXAMINER IS LOOKING FOR**

- **Classification:** invasive versus non-invasive.
- **Mucor versus Aspergillus** hyphae.
- **Fungal ball:** CT and treatment without antifungals.
- **Allergic fungal sinusitis:** Bent and Kuhn criteria, mucin, double density.
- **Mucormycosis:** risk factors, black eschar, amphotericin B and debridement.

## 7.2 Common pitfalls

**COMMON PITFALLS**

- Treating a diabetic with facial numbness and a black turbinate as simple sinusitis.
- Waiting for culture before starting amphotericin.
- Giving antifungals for a fungal ball, or omitting steroids in allergic fungal disease.

Written for this app from Oxford Textbook of Otolaryngology ch. 38, Oxford Specialist Handbook ch. 21, Dhingra ch. 37 and K. J. Lee ch. 30. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

IMAGING

# On imaging

*The imaging of this topic: key findings, differentials and representative scans.*

PARANASAL SINUSES · INFECTION & INFLAMMATION

## Noninvasive Fungal Sinusitis

### IMAGING FINDINGS

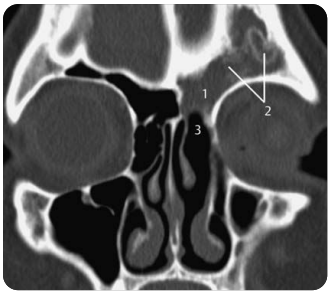
- Fungal ball: single sinus with dense contents and calcification, sclerotic walls
- Allergic fungal: expanded sinuses with central hyperdensity and low-density rim
- Very dark on T2

### TOP DIFFERENTIALS

- Polyposis
- Mucocele
- Inspissated secretions

### CHECKLIST

- Lamina papyracea and skull base remodelling
- Usually unilateral or asymmetric




Patient with constant left frontal sinus pressure. CT, coronal (a), sagittal (b), and axial ©. There is complete opacification of the left frontal sinus…



Axial NECT shows unilateral involvement of allergic fungal sinusitis. The involved sinuses are expanded, with centrally dense inspissated…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 700.

ORBIT & SINUS RELATIONS · INFECTION & INFLAMMATION

# Invasive Fungal Sinusitis

## IMAGING FINDINGS

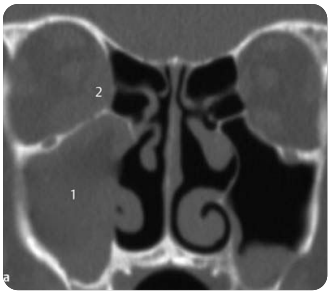
- Unilateral mucosal thickening with fat stranding outside the sinus
- Subtle bone erosion; loss of fat in the pterygopalatine fossa or orbit
- Non-enhancing mucosa ("black turbinate") on MRI

## TOP DIFFERENTIALS

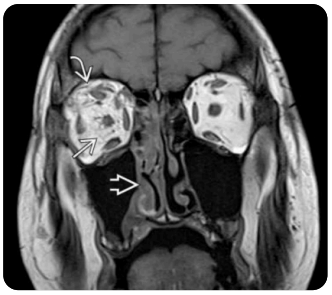
- Acute bacterial sinusitis with complication
- Lymphoma or carcinoma

## CHECKLIST

- Immunocompromised patient: endoscopy and biopsy without delay
- Orbit, cavernous sinus and brain on MRI




CT, coronal, bone-tissue setting. The bone-tissue setting reveals complete opacification of the right maxillary sinus (1), as well as very doubtful…



Coronal T1WI MR in patient with chronic myelogenous leukemia complicated by Rhizopus acute invasive fungal rhinosinusitis…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 704.